Journal of Ethnopharmacology 180 (2016) 28–36

Contents lists available at ScienceDirect

Journal of Ethnopharmacology

journal homepage: www.elsevier.com/locate/jep

The effects of Tao-Hong-Si-Wu on hepatic necroinflammatory activity and fibrosis in a murine model of chronic liver disease

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ARTICLE INFO

Article history:
Received 2 July 2015
Received in revised form
14 January 2016
Accepted 19 January 2016
Available online 20 January 2016

Keywords:
Tao-Hong-Si-Wu decoction
Hepatic necroinflammatory activity
Hepatic fibrosis
Hepatic function
Vascular endothelial growth factor
Flt-1
Kinase insert domain-containing receptor
Akt
Phosphorylated Akt

ABSTRACT

Background: Tao-Hong-Si-Wu decoction (THSWD) is a traditional Chinese herbal medicine that has been used for centuries in the treatment of Chinese patients with chronic liver disease. Recently, THSWD has been reported to alter vascular endothelial growth factor (VEGF) induced angiogenesis, raising the possibility that in addition to its anti-inflammatory properties; THSWD might also inhibit hepatic blood flow associated fibrosis.
Aim: To document the effects of THSWD on hepatic necroinflammatory disease activity, fibrosis and VEGF signaling in a murine model of chronic liver disease.
Methods: Sixty adult mice were equally divided into six study groups. Five groups were exposed to subcutaneous carbon tetrachloride (0.1 ml/10 g BW) for six weeks. Three of the five groups were treated with different concentrations of THSWD (4.25, 8.50, 17.00 g/kg), one with 0.1 mg/kg of Colchicine (positive control), and one with physiologic saline (negative control). Mice in the sixth group were not exposed to CCl_4 and remained untreated (healthy controls). Liver enzymes/function tests, hyaluronic acid and laminin levels were measured in serum, and hepatic histology, VEGF, Flt-1 and kinase insert domain-containing receptor (KDR), Akt and phosphorylated Akt (pAkt) expression were documented in liver tissue at the end of treatment.
Results: Hepatic necroinflammatory disease activity and fibrosis were significantly attenuated in THSWD treated mice in a dose dependent manner. These beneficial results were similar and often exceeded those achieved with Colchicine. In addition, VEGF, Flt-1, KDR, Akt and pAkt mRNA and protein expression were reduced in TSHWD treated mice.
Conclusions: In this animal model of chronic liver disease, THSWD decreased hepatic necroinflammatory disease and fibrosis. Inhibition of VEGF expression and downstream signaling were associated with these findings. Further studies with this and other TCMs as treatment for chronic liver disease are warranted.

1. Introduction

Left untreated, chronic hepatitis will often result in hepatic fibrosis and ultimately cirrhosis. Unfortunately, not all the causes of chronic hepatitis are treatable and in many conditions where specific treatments are available, responses rates can be suboptimal (Thabut et al., 2011; Song and Wang, 2008). In such settings, 'nonspecific' treatment of hepatitis and/or fibrogenesis is warranted (Rockey, 2013; Cohen-Naftaly and Friedman, 2011).

Traditional Chinese Medicine (TCM) consists of a complex

Abbreviations: THSWD, Tao-Hong-Si-Wu decoction; TCM, traditional Chinese medicine; CCl_4, carbon tetrachloride; VEGF, vascular endothelial growth factor; KDR, kinase insert domain-containing receptor; AST, aspartate aminotransferase; ALT, alanine aminotransferase; Alb, albumin; TBIL, total bilirubin; HA, hyaluronic acid; LN, laminin; ANOVA, analysis of variance; ELISA, enzyme-linked immunosorbent assay; HE, hematoxylin and eosin

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http://dx.doi.org/10.1016/j.jep.2016.01.030
0378-8741/

combination of natural herbs, minerals and/or animal products, each of which contains various chemical compounds. Together, these compounds can target different features of the disease process. In the setting of chronic liver disease, two such targets would be necroinflammatory disease activity (as reflected by serum liver enzyme/function tests and histology) and hepatic fibrosis (as reflected by serum hyaluronic acid (HA)/laminin levels and histology).

Recently, a significant component of hepatic fibrogenesis has been ascribed to increased levels of hepatic vascular endothelial growth factor (VEGF) and activation of its Flt-1 receptor and kinase insert domain-containing receptor (KDR) (Kantari-Mimoun et al., 2015; Yang et al., 2012; Ehling et al., 2014; Xu et al., 2005; Deli et al., 2005). This activation in turn results in increased Akt, phosphorylated Akt (pAkt) expression and ultimately, fibrogenesis. Thus, the expression of these markers would provide insights as to the efficacy and mechanism(s) whereby TCM might be of value in the treatment of chronic hepatitis (Wu et al., 2013a, 2013b; Zhao et al., 2014; Chen et al., 2015).

Tao-Hong-Si-Wu Decoction (THSWD) is composed of Semen Persicae, Flos Carthami, Radix Rehmanniae, Radix Angelicae Sinensis, Radix Paeoniae Rubra and Rhizoma Chuanxiong. THSWD was originally described in Detailed Explanation of the Jade Pivot (*Yu Ji Wei Yi*), a medical text authored by Xu Yan-Chun in the Ming dynasty. THSWD activates blood circulation, and inhibits inflammation (Jin et al., 2010; Li et al., 2010; Xie and Feng, 1986). It is widely applied to treat acute and chronic hepatitis (Chen et al., 1999; Li and Li, 2009; Wang, 2014; Lian et al., 2010; and Li and Li, 2013). However, the mechanisms whereby THSWD is of value in the treatment of these disease conditions has not been fully elucidated.

The present study was designed to document the effects of THSWD on hepatic inflammation and fibrosis in a commonly employed animal model of chronic liver disease. This study might provide some insights as to the possible mechanisms of action of THSWD.

2. Materials and methods

2.1. Experimental animals

Sixty 4 week old Kunming mice of both genders (30/30) weighing 16.0 ± 2.0 g were purchased from the SLAC Laboratory Animal Co. Ltd in Shanghai, China [License No. SCXK (Hu) 2012-0002]. The mice were bred in a specific pathogen-free facility in the Experimental Animal Center of Xiamen University [License No. SYXK (Min) 2013-0006] and allowed free access to food and water.

2.2. Experimental drugs

THSWD which is composed of Semen Persicae, Flos Carthami, Radix Rehmanniae, Radix Angelicae Sinensis, Radix Paeoniae Rubra and Rhizoma Chuanxiong (See Table 1), was purchased from the Xiamen Yanlaifu Pharmaceutical Co., Ltd. (Xiamen, China). Each herb was identified by Dr. Qiu Y. of the Pharmacy College of Xiamen University and together, were equivalent to 51 g of crude drug. All voucher specimens were deposited in the Chinese Medicine Research Centre of Xiamen University for future reference (No. 140218, 131125, 131129, 140211, 131209, 140117). The quality of these herbs was strictly controlled and processed according to the *Chinese Pharmacopoeia* (2010). The chemical compositions of all medicinal plants used in THSWD have been analyzed in previous experiments (Li et al., 2010), and the present experiments were conducted based on the results. Colchicine, 0.5 mg per tablet, (China SFDA: H53021369, product lot No. 131126) was purchased from the Xishuangbanna Pharmaceutical Co., Ltd. (Jinghong, China) and Carbon Tetrachloride (CCl_4), 500 ml per bottle (product lot No. 20140102) from the Dahao Fine Chemicla Co., Ltd. (Shantou, China).

2.3. Main reagents

Serum Alanine Aminotransferase Assay Kit, product lot No. Z403246, Aspartate Aminotransferase Assay Kit, product lot No. Z403249, Albumin Assay Kit, product lot No. Z401228 and Total Bilirubin Assay Kit, product lot No. Z403233 were purchased from the Beckman Coulter Experiment System (Suzhou) Co., Ltd. (Suzhou, China). A serum Laminin Quantitative Estimation Kit, product lot No. 20140624, and Hyaluronic Acid Quantitative Estimation Kit, product lot No. 20140606, were purchased from Auto Biological Engineering Corporation. CSB-E04756m Mice VEGF ELISA Kit, product lot No. U13030187, were purchased from Cusabio Biotech Co., Ltd. (Wuhan, China). 13687-1-AP VEGFR-1 (KDR) Antibody, product lot No. 00004813, was purchased from Proteintech Group, Inc. (Wuhan, China). BS1373 VEGFR2 (Flt-1) (F945) Polyclonal Antibody, product lot No. CJ36131, BS1978 AKT (E40) Polyclonal Antibody, product lot No. CJ36132, and BS4007 p-Akt (S473) Polyclonal Antibody, product lot No. CJ00101, were purchased from Bioworld Technology, Inc. (Minnesota, USA).

2.4. Instruments

The following instruments were employed: DXC800 Beckman automatic biochemistry analyzer (Beckman coulter, inc., Brea, California, USA), Olympus CX41 microscope (Olympus optical co. ltd, Tokyo, japan), LeicaRM2016 Histotome (LEICA co., Solms, Germany), TDZ5-WS low-speed Multi-pipe carriers Autobalance centrifuge (Xiang Yi centrifuge instrument co., ltd., Changsha, china), 9600 PCR Amplification system (applied Biosystems inc., foster city, California, USA), Typ-nr. B337874 semi-dry transfer unit (Biometra GmbH, Göttingen, Germany), TS-8 transference Decoloring shaker (Haimen Kylin-bell lab instruments co., ltd., Haimen, china), GNP-9080-water-insulation Thermostatic incubator (shanghai Jing Hong laboratory instrument co., ltd., shanghai, china), Multiskan MK3 Multimode reader (Thermo fisher scientific inc., Waltham, MA, USA), DYY-8 steady flow electrophoresis apparatus (shanghai Qite analytical instrument co., ltd., shanghai, china), H6-1 Microelectrophoresis chamber (shanghai Jingyi organic glass instrument factory, shanghai, china), FR-980A gel

Table 1
Contents of Tao-Hong-Si-Wu Decoction (THSWD).

Chinese name	Botanical name	Common name	Weight (g)	Part used	Voucher Numbers
Tao Ren	*Prunus persica* (L.) Batsch or *Prusua davidiana* (Carr.) Franch.	Semen Persicae	9	Seed	140218
Hong Hua	*Carthamus tinctorius* L.	Flos Carthami	6	Flower	131125
Sheng Di Huang	*Rehmannia glutinosa* Libosch.	Radix Rehmanniae	12	Root tuber	131129
Dang Gui	*Angelica sinensis* (Oliv.) Diels.	Radix Angelicae Sinensis	9	Root	140211
Chi Shao	*Paeonia lactiflora* Pall. or *Paeonia veitchii* Lynch	Radix Paeoniae Rubra	9	Root	131209
Chuan Xiong	*Ligusticum chuanxiong* Hort.	Rhizoma Chuanxiong	6	Rhizome	140117

image analysis system (shanghai Furi technology co., ltd., shanghai, china), TU-1901 ultraviolet spectrophotometer (Beijing Purkinje general instrument co., ltd., Beijing, china) and ONE-DScan program (Scanalytics inc., Fairfax, Va., USA).

2.5. Medicinal preparation

THSWD was dissolved in distilled water (1:8) for 20 min, decocted for 30 min, and filtered with 8 layers carbasus; the decocted gruffs were again decocted in water (1:6) for 30 min, and filtered with 8 layers carbasus. The decocted, evaporated and concentrated decoction was then suspended at concentrations of 1.70 g/ml, 0.85 g/ml and 0.425 g/ml. The remainder of the decoction was lyophilized into powder and preserved at 4 °C for subsequent use. Colchicine was prepared with double steamed water to a concentration of 0.01 mg/ml.

2.6. Animal groupings, modeling and drug administration

Sixty mice were divided into 6 groups ($n=10$/group) by random digits table. Of the five groups exposed to $CC1_4$, three were treated with either high (1.7 g/ml), middle (0.85 g/ml) or low (0.425 g/ml) dose THSWD (THSWD-H, M, L respectively), one with Colchicine (positive control) and one with physiologic saline (untreated control). The THSWD-H, THSWD-M and THSWD-L treated groups received their treatments by intragastric administration of 0.2 ml/10 g THSWD aqueous solution once a day. The doses of THSWD were equivalent to 20, 10 and 5 times of human adult dose, respectively. The Colchicine treated group was treated with intragastric administration (0.2 ml/10 g) of Colchicine (0.1 mg/kg) once a day for 6 weeks. The dose of Colchicine was equivalent to dose given to human adult liver disease patients. Untreated controls received physiological saline by the same route. In these groups, 40% $CC1_4$ was injected subcutaneously at a dose of 0.1 ml/10 g with Arachis Oil every five days for 6 weeks. The dosage was doubled on the first administration. A sixth group of non-$CC1_4$ exposed mice (healthy controls) were injected with 0.1 ml/10 g Sodium Chloride Physiological Solution in an identical manner.

2.7. Serum AST, ALT, alb, TBIL, HA and LN determinations

Twenty-four hours after the last administration of $CC1_4$, mice were anesthetized by inhaling ethylether, after which 0.8 ml of peripheral blood was collected from the orbital sinus of each mouse prior to sacrifice. The death procedure by cervical dislocation after etherization was approved (No. XMULAC 2012-0039) by the Laboratory Animal Administration and Ethics Committee of Xiamen University. Serum ALT, AST, Albumin and TBIL levels were detected using commercial kits and a supermatic biochemistry analyzer as per the manufacturer's instructions. Serum HA and LN determinations required 50 μl of serum (for detecting HA, 50 μl of the binding protein solutions was required) being vibrated for 30 seconds, incubated for 30 min at 37 °C, washed 5 times and the addition of 100 μl luminescent substrate (added away from light) for 5 min at room temperature prior to being detected by a supermatic chemiluminescence immune analyzer as per the manufacturer's instructions.

2.8. Histology

At the time of sacrifice, portions of liver tissue were removed and fixed in a 10% neutralized formaldehyde solution. Paraffin sections (4μm) were developed and stained with Haematoxylin and Eosin. All sections were prepared for routine histological evaluations. Hepatic histologic changes were observed by light microscopy and recorded by photography ($\times$ 100). Analyses of the histology was performed by an individual unaware of the treatment groupings.

2.9. VEGF expression in hepatic tissues

At the time of sacrifice, portions of liver tissue not required for histology were cut into fine pieces by ophthalmic scissors, transferred to a homogenizer, exposed to protein extracting solution, homogenated under iced bath, transferred to a centrifuge tube and centrifuged at 15000 r min^{-1} for 30 min. The precipitate was discarded and clear supernatant stored at -20 °C prior to analysis. A vascular endothelial growth factor (VEGF) ELISA Kit was used to determine the concentrations of tissue VEGF according to the manufacturer's instructions. Briefly, 100 μl of sample was incubated with 100 μl of VEGF biotin labelled antibody for 60 min at 37 °C followed by 3 washes. 100 μl of horseradish peroxidase (HRP) labelled avidin fluid was then added and incubated for 30 min at 37 °C followed by 5 washes. Finally, 90 μl of substrate solution was added and incubated for 25 min at 37 °C before applying 50 μl of stop solution. Optical density (OD) values were read with a Multimode Reader at 450 nm wavelength.

2.10. Flt-1, KDR, Akt and pAkt protein expression in hepatic tissues

Liver tissue was placed in 95% physiological saline, rinsed several times and transferred to a homogenizer containing RIPA buffer lysate. Protein concentrations were measured using the Bradford colorimetric method. Denatured protein was resolved by 8% SDS-polyacrylamide gel electrophoresis (SDS-PAGE) in an electrophoresis chamber and transferred to polyvinylidene difluoride membranes. Blots of Flt-1, KDR and Akt were blocked with 5% non-fat dry milk-TBST buffer. The pAkt blot was blocked with BS4007 p-Akt (S473). Membranes were incubated overnight at 4 °C with 1:1000 dilution of antibodies for Flt-1, KDR, Akt and pAkt. Equal lane loading was assessed using B-actin. Blots were rinsed 5 times with TBST buffer for 3 min each. Washed blots were incubated with 1:10000 dilution of the horseradish peroxidase conjugated-secondary antibody for 1 h and washed 5 times with TBST buffer. The transferred proteins were visualized with luminescence fluid and X-ray fixing. Films were scanned, and the average optical density of the blots was analyzed with an Image-Pro Plus 6.0 image processing system.

2.11. Reverse transcription polymerase chain reaction (RT-PCR) measurements of VEGF, Flt-1, KDR and Akt mRNA in hepatic tissues

Total RNAs were extracted from hepatic tissue (100 mg) using a trizol reagent according to the manufacture's instruction. RNAs were dissolved in RNase-free water and the concentration determined spectrophotometrically. cDNA was synthesized by reverse transcription from 1 μg of total RNA using reverse transcription system and oligo dT15 as the primer to a final volume of 20 μl. The reverse transcription products were diluted into 100 μl as cDNA templates. The PCR reaction was performed on a PCR Amplification System in a total volume of 25 μl containing 1 μl of cDNA templates, 0.5 μl (10 μm) primers and Go Taq Green Master Mix 12.5 μl. The primer sequences were as follows: GAPDH, forward 5′-ATCATGTTTGAGACCTTCAACA-3′, reverse 5′-CATCTCTTGCTCGAAGTCCA-3′ VEGF, forward 5′-TTCAGAGCGGAGAAAGCATT-3′, reverse 5′-GAGGAGGCTCCTTCCTGC-3′ Flt-1, forward 5′-GGAGGGATAACAGGCAAT-3′, reverse 5′-CCATCAGGGGTAAGAGTA-3′ KDR, forward 5′-TCTGTGGTTCTGCGTGGAGA-3′, reverse 5′-GTATCATTTCCAACCACCCT-3′ Akt, forward 5′-CGGATACCATGAACGACGTAG-3′, reverse 5′-GCAGGCAGCGGATGATAAAG-3′. The mRNA was specified by the primer sequences. PCR was performed for 35 cycles. Each cycle for GAPDH, VEGF, KDR and

Akt consisted of predenaturation for 2 min at 94 °C, denaturation for 30 s at 94 °C, annealing for 30 s at 53 °C and extension for 30 s at 72 °C. Each cycle for Flt-1 consisted of predenaturation for 2 min at 94 °C, denaturation for 30 s at 94 °C, annealing for 30 s at 48 °C and extension for 30 s at 72 °C. Each PCR product was subjected to electrophoresis on 2% agarose gel. Ethidium bromide-stained bands were scanned and analyzed by using the ONE-DScan program.

2.12. Statistical analysis

Parametric data were expressed as mean ± SD ($\bar{x} \pm s$) and analyzed with One-Way ANOVA (analysis of variance) by the Statistical Product and Service Solutions (SPSS) 19.0 (IBM, Armonk, NY, USA). The least significant difference (LSD) method was used for individual comparisons. A P value of <0.05 was regarded as statistically significant.

3. Results

3.1. Effects of THSWD on hepatic necroinflammatory disease activity

To determine if THSWD attenuates hepatic necroinflammatory disease activity in this $CC1_4$-induced model of liver disease, serum ALT, AST, Alb, TBIL and hepatic histology were documented at the end of the study. Colchicine was used as the positive control. As shown in Fig. 1, serum ALT levels were significantly lower in the

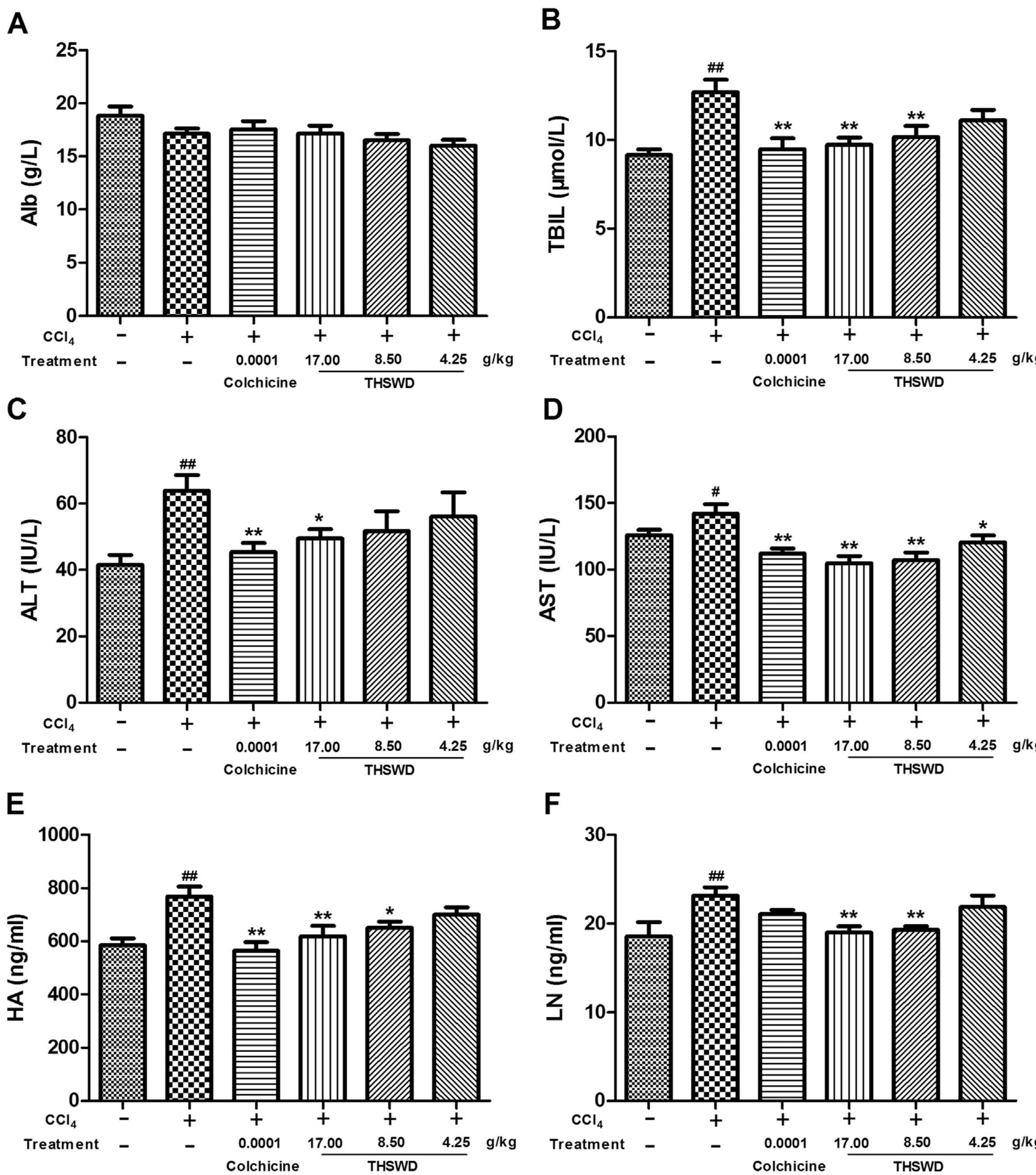

Fig. 1. Effects of THSWD on hepatic function and fibrosis correlated factors in a CCl_4-induced liver injury model. A shows effect of THSWD on albumin (Alb), B shows effect of THSWD on total bilirubin (TBIL), C shows effect of THSWD on alanine transarninase (ALT), D shows effect of THSWD on aspartate aminotransferase (AST), E shows effect of THSWD on hyaluronic acid (HA) and F shows effect of THSWD on laminin (LN). Data were represented as mean ± SD (n=6 or 7 mice per group). $^{\#}P<0.05$, $^{\#\#}P<0.01$ compared with healthy controls; $^{*}P<0.05$, $^{**}P<0.01$ compared with saline treated controls.

THSWD-H treated group, AST in all THSWD treated groups and TBIL in THSWD-H and THSWD-M treated groups. In the case of AST and TBIL, the results with THSWD-H were similar or greater than those achieved with Colchicine. Serum albumin levels remained unchanged in all $CC1_4$ exposed groups when compared to healthy controls.

In terms of histology, as expected, the lobular architecture of the liver in healthy controls was intact with well-arranged hepatic cords, normal hepatic sinusoids and no inflammatory cell infiltration. However, in untreated $CC1_4$-exposed mice, there was lobular disarray, irregular hepatic cell cords, hepatocellular swelling, fatty degeneration and zone I inflammatory cell infiltrates. When compared to untreated controls, these changes were less prominent in the THSWD and Colchicine treated groups (Fig. 2). These data suggest that THSWD inhibits $CC1_4$-induced hepatic necroinflammatory disease.

3.2. Effects of THSWD on hepatic fibrosis

To evaluate if THSWD inhibits $CC1_4$-induced hepatic fibrosis, serum HA and LN levels were determined and hepatic histology analyzed at the end of treatment. As shown in Fig. 1, serum HA and LN levels in THSWD-H and THSWD-M treated groups were significantly lower than those of untreated controls ($P<0.05$, $P<0.01$). Indeed, the effects of THSWD-H and THSWD-M on LN levels were more pronounced than those achieved with Colchicine and similar to healthy control mice.

The results of histologic analysis for hepatic fibrosis are presented in Fig. 2. Here, collagen was less prominent in THSWD and Colchicine treated groups compared to untreated controls. These results suggest that THSWD can attenuate hepatic fibrosis in this model of chronic liver disease.

3.3. Effects of THSWD on hepatic VEGF expression

Pathological angiogenesis occurs differently from vascular repair and negatively affects hepatic fibrosis (Folkman, 1995; Coulon et al., 2011). Thus we measured VEGF in hepatic tissue. ELISA analyses of liver tissues revealed that treatment with all three concentrations of THSWD decreases VEGF protein when compared to untreated controls. VEGF mRNA expression was also decreased in the THSWD treated groups. Indeed, in the THSWD-H treated group, VEGF mRNA levels were lower than those of the Colchicine, untreated and healthy controls (Figs. 3A and 4 A respectively).

3.4. Effects of THSWD on hepatic VEGF, Flt-1, KDR and Akt mRNA expression

In order to further define the mechanism whereby THSWD reduces the pathological changes caused by $CC1_4$, we documented the gene transcription of PI3K-Akt associated factors such as Flt-1, KDR and Akt in hepatic tissue of mice with $CC1_4$-induced liver disease. In terms of mRNA expression, the results of RT-PCR are shown in Figs. 3B and C, and 4A–C. THSWD-H treatment was associated with significantly lower Flt-1, KDR and Akt mRNA expression, more so than that of Colchicine (Fig. 4). THSWD-M treatment was associated with lower Flt-1 and KDR mRNA expression while THSWD-L lowered only Flt-1 mRNA expression.

3.5. Effects of THSWD on hepatic Flt-1, KDR, Akt and pAkt protein expression

VEGF binding to its receptors activates the PI3K-Akt pathway which regulates hepatic fibrosis (Byeon et al., 2010; Zhang et al., 2014; Huang et al., 2011). Thus, the effects of THSWD on this signaling pathway were documented. Western blot results indicated that THSWD-H significantly reduced Flt-1, KDR, Akt and pAkt protein expression ($P<0.05$, $P<0.01$) (Fig. 5). THSWD-M decreased KDR and Akt protein levels while THSWD-L had no effect on these makers. Together with the results of mRNA expression, these results suggest that P13K-Akt signaling is inhibited by decreasing VEGF, Flt-1, Akt and pAkt expression.

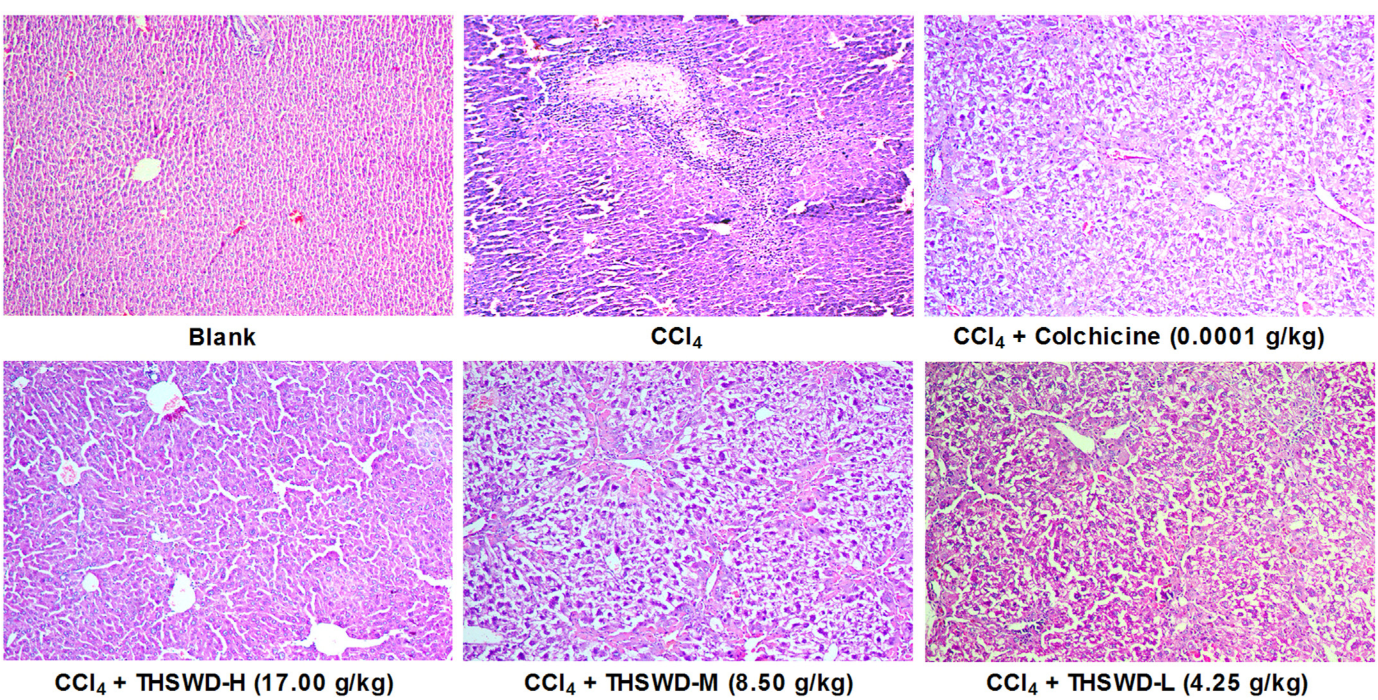

Fig. 2. Effects of THSWD on hepatic necroinflammation and fibrosis in a CCl_4-induced liver injury model. Sections were stained with hematoxylin and eosin (H&E) and viewed at a magnification of $\times$ 100.

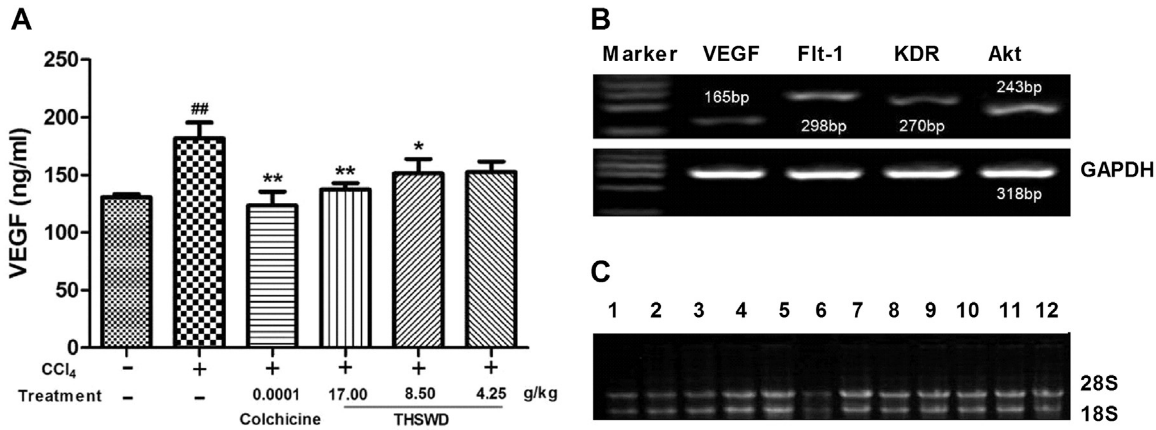

Fig. 3. A: Effects of THSWD on VEGF protein levels in a CCl_4-induced liver injury model. Data are presented as mean ± SD (n=6 mice per group). $^{\#}P<0.05$, $^{\#\#}P<0.01$ compared with healthy controls; $^{*}P<0.05$, $^{**}P<0.01$ compared with saline treated controls. B: Gel electrophotogram of target gene VEGF, Flt-1, KDR and Akt amplification products of reverse transcription polymerase chain reaction. C: Gel electrophotogram of total RNA of hepatic tissue. Two samples from each group were randomly selected for integrity verification. 1–2: healthy controls; 3–4: saline treated controls; 5–6: Colchicine; 7–8: THSWD (17.00 g/kg); 9–10: THSWD (8.50 g/kg); 11–12: THSWD (4.25 g/kg).

4. Discussion

The results of this study indicate that THSWD, a traditional Chinese Herbal medication, decreases serum biochemical and histologic evidence of $CC1_4$-induced hepatic necroinflammation. The results also indicate that hepatic fibrosis is attenuated in this model of chronic liver disease. Although the precise mechanisms have yet to be defined, significant decreases in VEGF and VEGF signaling appear to be involved.

A number of previous reports have described TCMs as being effective in preventing and/or decreasing hepatic necroinflammation in animal models of liver disease. For example, Wang et al. described lower ALT, AST and less histologic evidence of inflammation with the TCM Zhi-Zi-Da-Huang which was

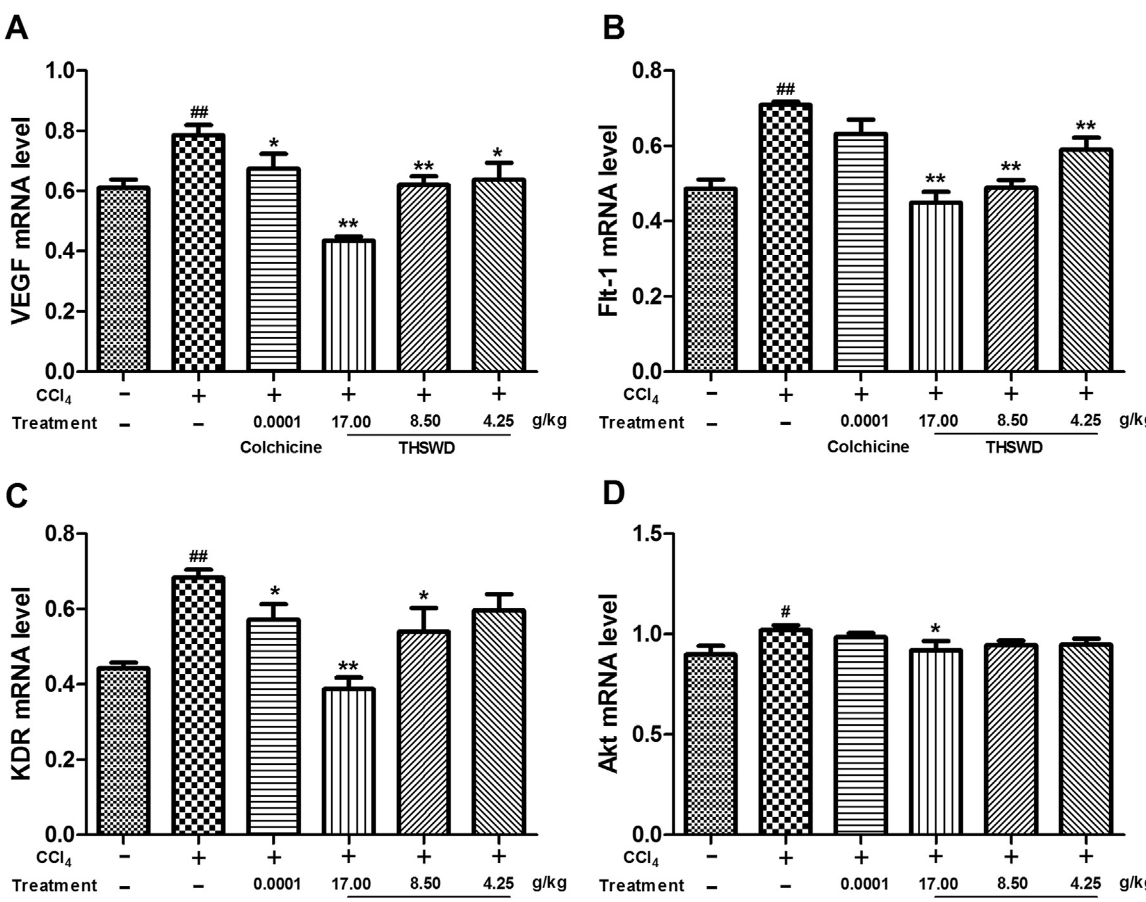

Fig. 4. Effects of THSWD on mRNA expression of VEGF, Flt-1, KDR and Akt in a CCl_4-induced liver injury model. Data were represented as mean ± SD (n=6 mice per group). $^{\#}P<0.05$, $^{\#\#}P<0.01$ compared with healthy controls; $^{*}P<0.05$, $^{**}P<0.01$ compared with saline treated controls.

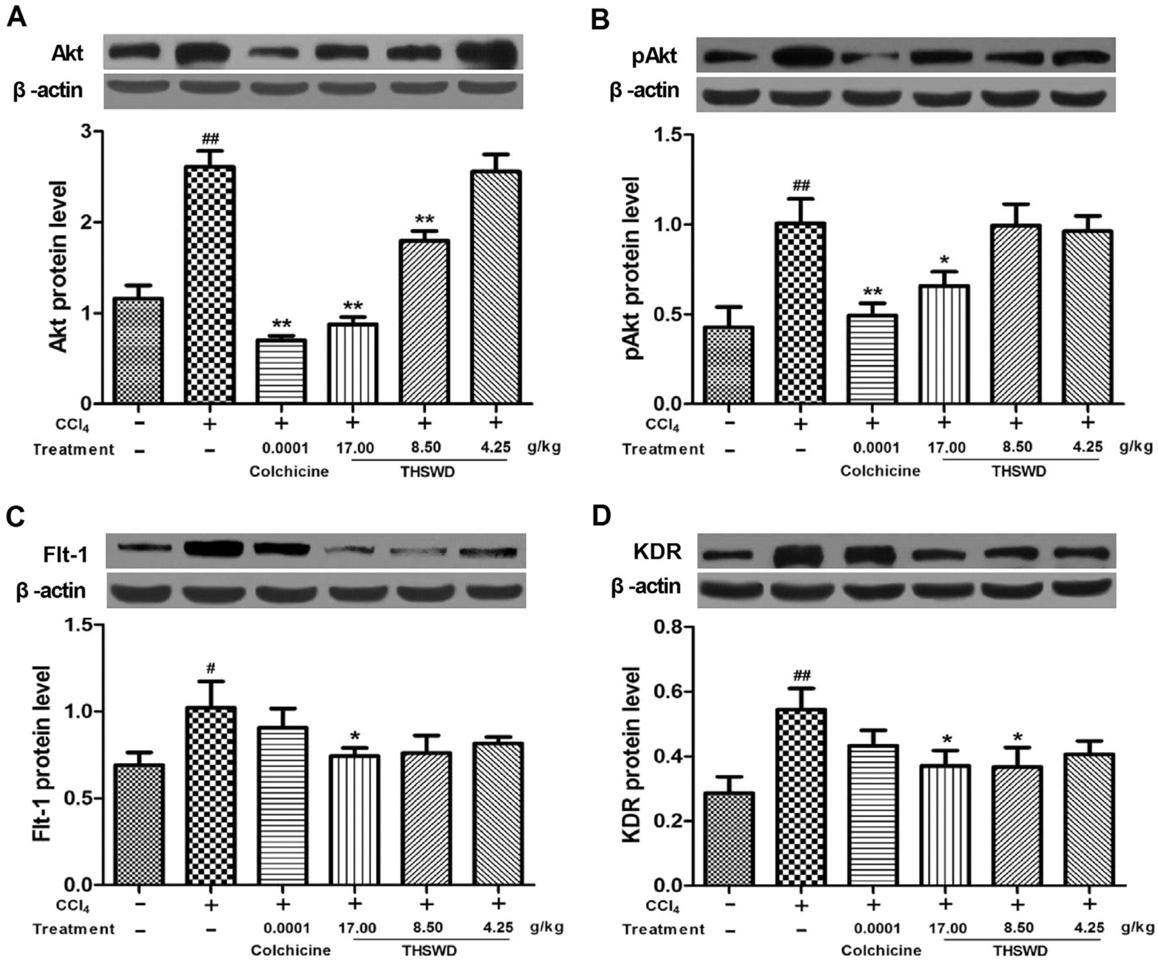

Fig. 5. Effects of THSWD on protein expression of Akt, pAkt, Flt-1 and KDR in hepatic tissue of CCl_4-induced liver injury model. Data were represented as mean ± SD (n=6 mice per group). $^{\#}P<0.05$, $^{\#\#}P<0.01$ compared with healthy controls; $^{*}P<0.05$, $^{**}P<0.01$ compared with saline treated controls.

administered to rats with alcohol induced liver disease (Wang et al., 2009). Wu et al. reported similar benefits with Da-Huang-Fu-Zi-Tang in rats exposed to toxic concentrations of sodium taurocholate (Wu et al., 2013a, 2013b). In yet another model, Yu-langsan polysaccharide pre-treatment resulted in significant biochemical and histologic improvement in mice exposed to the potent hepatotoxin nimesulide (Nguyen et al., 2015). Thus, our findings are in keeping with these previous reports and extend the benefits of TCMs to include the more commonly employed, $CC1_4$-induced model of liver injury.

The mechanism(s) whereby THSWD inhibited hepatic necroinflammatory disease activity remains to be determined. Although oxidative stress and anti-oxidant activity were not documented in the present study, a significant decrease in free oxygen radicles and reactive oxygen species has been described with other TCMs. Moreover, THSWD contains the herb *Carthamus tinctorius* L. which has been previously documented to possess potent anti-oxidant properties (Wu et al., 2013a, 2013b). None-the-less, additional studies are required to identify the precise mechanisms involved.

In contrast to the abundance of reports describing the beneficial effects of TCMs on hepatic necroinflammatory disease, relatively few reports have been published in the English literature describing the effects of TCMs on hepatic fibrosis and to our knowledge, only several have focused specifically on $CC1_4$-induced cirrhosis. In the first, Zhang et al. described lower serum hyaluronic acid concentrations, hepatic hydroxyproline and proline levels and decreased hepatic fibrosis in $CC1_4$-exposed rats treated with an in house TCM formulation (Zhang et al., 1996). In their study, TCM was more effective than hepatic stimulator substance, a combination of selenium plus Vitamin E and ciprofloxacin in decreasing hepatic fibrosis. In the other previously published report, Lin et al. administered Fufang-Liu-Yue-Qing to rats exposed to $CC1_4$ for 12 weeks. Compared to untreated controls, liver enzymes, pro-inflammatory cytokine levels, α-SMA, TGF-β_1 and collagen deposition were significantly reduced in TCM treated animals. In addition, the authors reported enhanced extracellular matrix degradation and enhanced hepatic stellate cell apoptosis with their TCM (Lin et al., 2012). In neither of the above studies, did the authors attempt to determine whether the decreased fibrosis resulted from decreased fibrogenesis or increased fibrinolysis.

The distinction between decreased fibrogenesis and increased fibrinolysis is an important one in view of our findings that THSWD significantly decreases VEGF levels and downstream signaling. Specifically, recent data from Kantari-Mimoun et al. and others indicate that increased VEGF activity enhances hepatic fibrogenesis during the development of cirrhosis by increasing neovascularization in the peri-fibrosis region and increasing endothelial cell permeability. On the other hand, once cirrhosis is established and the source of chronic injury has resolved, VEGF enhances fibrinolytic activity by inducing revascularization of established fibrous tissue (Kantari-Mimoun et al., 2015; Yang et al., 2012; Xu et al., 2005). Thus, if future studies confirm our findings, the administration of THSWD should be limited to the early phase of liver injury when fibrogenesis is still occurring and increased

VEGF activity is deleterious rather than when extensive fibrosis or cirrhosis are already established and increased VEGF activity is beneficial.

There are a number of limitations to the present study that warrant emphasis. First, as is often the case with TCMs, it remains unclear as to whether one ingredient or a combination of ingredients was responsible for the beneficial effects observed. Second, as noted earlier, the mechanism whereby THSWD attenuated hepatic necroinflammatory disease activity was not studied and in the case of hepatic fibrosis, it remains unclear as to whether decreased fibrogenesis or increased fibrinolysis were responsible for this finding. Third, whether the decrease in VEGF expression described reflects less “scar associated monocyte”, macrophage or sinusoidal endothelial cell expression of VEGF remains to be determined. Finally, although encouraging, the beneficial effects of THSWD in this animal model of liver injury will have to be demonstrated in prospective, controlled trials in humans prior to being considered as a therapeutic option for patients with chronic liver disease.

5. Conclusion

In summary, THSWD decreased hepatic necroinflammatory disease and fibrosis in a $CC1_4$-induced model of liver injury. It also decreased VEGF expression and down-stream signaling in this model. Although those findings need to be confirmed, they serve to underscore the potential value of TCMs in the treatment of chronic liver disease and encourage further research into this therapeutic modality.

Author contributions

Xi, S.Y., Wang, Y.H. and Gong, Y.W. participated in the study design evaluated the statistical analysis. Xi, S.Y., Minuk, G.Y. and Xu, Y.X.Z. wrote the paper and critically revised the manuscript. Shi, M.M., Peng, Y., Cheng, Y. and Wang X.R. carried out the experiment. Jiang, X.Q., Yang, J.Q. and Yue, L.F. performed the statistical analysis. Xi, S.Y. also involved in drafting the picture and the manuscript.

Conflicts of interest

All the authors declare no present or potential conflicts of interest. All authors are responsible for the content and writing of the paper and approved of its publication.

Acknowledgments

This work was supported by the National Natural Science Foundation of China (No. 81202659), the Natural Science Foundation of Fujian (China) (No. 2014J01373), the Scientific Research Start Foundation for New Teacher of Xiamen University (No. ZK1014), the Xiamen Science and technology Key Program Plan Grant (No. 3502Z20100006) and Fujian Undergraduate Training Programs for Innovation and Entrepreneurship (No. DC2013174). We gratefully thank Dr. Xi Huang of Nanjing University of Chinese Medicine for academic guidance, the experimentalist Dr. Fu Chen of the Medicine Laboratory Animals Center of Xiamen University for assisting in laboratory procedures, Dr. Sudharsana R. Ande, Dr. Pollock Galia and the secretary Roberta Vizniak of John Buhler Research Centre for revising this paper.

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